O-S-00650 (INT-Bu Mines)
February 25, 1965

INTERIM FEDERAL SPECIFICATION

SODIUM SULFITE, ANHYDROUS, TECHNICAL, AND SODIUM BISULFITE, ANHYDROUS, TECHNICAL, FOR WATER TREATMENT

This Interim Federal Specification was developed by the Bureau of Mines, College Park, Maryland, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers anhydrous sodium sulfite (Na_2SO_3) and anhydrous sodium bisulfite ($Na_2S_2O_5$), intended for use in the treatment of industrial water as designated.

1.2 Classification.

1.2.1 Sodium sulfite covered by this specification shall be furnished in one grade and one class.

1.2.2 Sodium bisulfite, also known as sodium pyrosulfite and anhydrous sodium metabisulfite, covered by this specification shall be furnished in one grade and in one class only.

2. APPLICABLE SPECIFICATIONS AND STANDARDS.

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

L-P-378 - Plastic film (polyethylene thin gage)
UU-S-48 - Sacks, Shipping
PPP-D-723 - Drums, Fiber

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Material. The sodium sulfite and sodium bisulfite shall be as specified hereinafter and suitable for the purpose intended (6.1).

3.2 The sodium sulfite and the sodium bisulfite shall be free from admixture of additive chemicals and shall conform to the chemical and physical properties specified in table I when tested in accordance with 4.4 and 4.5.

3.3 Particle size. The sodium sulfite and the sodium bisulfite shall be free from lumps and foreign matter and shall be a free-flowing powder satisfactory for use in a dry-feed machine.

3.3.1 100-percent shall pass through a No. 18 (1,000 micron) sieve.

Table I - Physical and chemical tolerances

Property	Limits, anhydrous sodium sulfite	Limits, anhydrous sodium bisulfite
Assay	96.0% Na_2SO_3, minimum	65.0% SO_2, minimum
Insoluble matter	0.2% , maximum	0.1% , maximum
Sodium chloride	1.0% , maximum	1.0% , maximum
Iron	0.1% , maximum	0.02% , maximum
Appearance	tan to pinkish powder	yellowish white crystals
Color (clear 10% solution)	200 APHA, maximum	100 APHA, maximum
Odor	very faint , maximum	not applicable

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. A representative composite sample shall be taken, placed in a clean, dry, glass container, sealed, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.3 Inspection and tests. Material offered for delivery under this specification, and packaging thereof, shall be subject to inspection and tests by Government inspectors.

4.4 Test procedures for sodium sulfite, anhydrous, technical.

4.4.1 General. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be conducted in parallel with each test, using the same quantities of reagents, and corrections shall be applied when significant. All weighings shall be made to four significant figures, unless otherwise specified.

4.4.2 Assay. Weigh (to three significant figures) a sample of about 0.3 g. and add it to 50.0 ml. of 0.1N iodine solution in a 250-ml. beaker or casserole (the sulfite must be added to the iodine solution). Let stand for 5 minutes to react (reaction may be hastened by crushing agglomerates but do not stir); then add 100 ml. water and 2 ml. conc. hydrochloric acid. Stir, then titrate (10 ml. burette, readable to 0.01 ml.) the excess iodine with 0.1N sodium thiosulfate, using starch as the indicator towards the end of the titration (light straw color).

4.4.2.1 Iodine solution. Dissolve 23 g. potassium iodide (KI) in 500 ml. of distilled water. Add 12.691 g. iodine (I_2) and stir to dissolve. Dilute to 1 liter and store in a dark glass bottle. Standardise against tenth normal thiosulfate which has been standardised against potassium dichromate (primary standard).

4.4.2.2 Sodium thiosulfate. Dissolve 16.0 g. sodium thiosulfate ($Na_2S_2O_3$) in distilled water and dilute to 1 liter. Add 1 g. sodium carbonate (Na_2CO_3) and 1 ml. chloroform ($CHCl_3$). Use potassium dichromate (primary standard) to standardise and adjust normality to 0.100 ± 0.005.

4.4.2.3 Calculate percent sodium sulfite as follows:

$$\text{Percent } Na_2SO_3 = \frac{6.303\ (50n_1 - an_2)}{W}$$

Where:

n_1 = normality of iodine solution
a = volume of sodium thiosulfate solution in ml.
n_2 = normality of sodium thiosulfate solution
W = weight of sample in g.

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4.4.3 Insoluble matter. Dissolve 10 g. of sample in 100 ml. of water and heat on a steam bath for 1 hour. Filter through asbestos-matted and tared Gooch crucible, wash thoroughly with hot water, dry in an oven at 105° C for 3 hours, and weigh.

4.4.3.1 Calculate the percentage of insoluble matter as follows:

$$\text{Percentage of insoluble matter} = \frac{100A}{B}$$

Where:

A = weight of residue in g.

B = weight of sample in g.

4.4.4 Sodium chloride.

4.4.4.1 Control solution. Dissolve 0.1000 g. of sodium chloride in 1 liter of water.

4.4.4.2 Procedure. Dissolve 1.000 g. of sample in 10 ml. of water. Add 2 ml. of 30-percent hydrogen peroxide, allow the solution to stand for 30 minutes, and then dilute to 100 ml. To 50 ml. add 2 ml. of 1:1 nitric acid and 1 ml. of a 3-percent aqueous solution of silver nitrate. Any resulting turbidity shall not be greater than that produced in an equal volume of the control solution (4.4.4.1) to which 2 ml. of 1:1 nitric acid and 1 ml. of 3-percent silver nitrate have been added.

4.4.5 Iron.

4.4.5.1 Control solution (0.01 g. Fe per liter). Dissolve 0.702 g. of ferrous ammonium sulfate hexahydrate ($Fe(NH_4)_2(SO_4)_2.6H_2O$) in 10 ml. of 10 percent sulfuric acid and dilute to 100 ml. with water. To 10 ml. of this solution add 10 ml. of 10-percent sulfuric acid and dilute to 1 liter.

4.4.5.2 Procedure. Dissolve 0.100 g. of the sample in 10 ml. of water and dilute to 100 ml. To 1 ml. of the solution add 2 ml. of hydrochloric acid and evaporate to dryness on a steam bath. Dissolve the residue in 10 ml. of water, add 2 ml. of hydrochloric acid, dilute to 50 ml. and then add 30 to 50 mg. of ammonium chloride crystals. Add 3 ml. of ammonium thiocyanate solution made by dissolving 75 g. of ammonium thiocyanate in 250 ml. of water. Any red color should not exceed that produced by 1 ml. of the control solution in an equal volume of solution containing the quantities of reagents used in the test.

4.4.6 Appearance. Examine the sample in the bottle and describe it in appropriate terms, such as white, light tan, dark brown powder, etc.

4.4.6.1 Note and record any unusual conditions, such as foreign matter, dirt or rust.

4.4.7 Color (clear 10% solution). Weigh 10 g. of the sample to the nearest tenth of a g. and transfer to a clean dry 250-ml. beaker. Add (graduate) exactly 90 ml. of distilled water and stir to dissolve the sample. Filter the solution through a No. 42 Whatman filter paper, repeating the filtration if the filtrate is not clear. Collect 50 ml. of the filtrate in a clean, dry, tall-form 50-ml. Nessler tube. Compare the color to standard 200 APHA color (Hazen); report "acceptable" only if the APHA color standard is not exceeded.

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4.4.8 Odor. If the bottle is full, pour out some of the sample to leave at least 1 inch of air space below the open end of the bottle. Stopper the bottle and invert 3 or 4 times to mix the contents. Allow the stoppered bottle to stand one minute, remove the cap and cautiously smell the vapors. Report the odor, if any, using one of these five levels of odor intensity: none (no odor perceptible); very faint (detectable by an experienced observer, but not by an average person); faint (detectable by the average person if his attention is called to it); distinct (readily detectable); strong (so intense that the material is unfit for use). Where applicable, supplement this grading if the odor is familiar by describing it as phenolic, sweetish, moldy, sour, fishy, etc.

4.4.9 Exceptions. When a contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government at its discretion may modify in whole or in part the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures spelled out in this specification.

4.5 Test procedure for sodium bisulfite, anhydrous, technical.

4.5.1 General. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be conducted in parallel with each test, using the same quantities of reagents, and corrections shall be applied when significant. All weighings shall be made to four significant figures, unless otherwise specified.

4.5.2 Assay. Weigh (to three significant figures) a sample of about 0.25 g., and then follow the procedure of 4.4.2.

4.5.2.1 Iodine solution. Determine in accordance with procedure outlined in 4.4.2.1.

4.5.2.2 Sodium thiosulfate. Determine in accordance with procedure in 4.4.2.2

4.5.2.3 Calculate percent sulfur dioxide (SO_2) as follows:

$$\text{Percent } SO_2 = \frac{3.203 \times (50n_1 - an_2)}{W}$$

Where:

n_1 = normality of iodine solution.

a = volume of sodium thiosulfate solution in ml.

n_2 = normality of sodium thiosulfate solution.

W = weight of sample in g.

4.5.3 Insoluble matter. Determine in accordance with 4.4.3.

4.5.4 Iron. Determine in accordance with 4.4.5.

4.5.5 Appearance. (see 4.4.6).

4.5.6 Color (clear 10% solution). Follow the directions in 4.4.7, but use a standard 100 APHA color (Hazen) to test for acceptability.

4.5.7 Exceptions. (see 4.4.9).

4.5.8 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packing and marking requirements comply with section 5 of this specification.

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5. PREPARATION FOR DELIVERY.

(For civil agency procurement the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packing. Packing shall be level A, B or C as specified (6.2).

5.1.1 Level A. The sodium sulfite or sodium bisulfite furnished in quantities of 100 pounds shall be packed in a close-fitting bag conforming to UU-B-48, grade 14I. When specified the sodium sulfite or sodium bisulfite furnished in quantities of 375 pounds shall be packed in drums conforming to PPP-D-723. A bag or liner conforming to L-P-378, type I, grade A, finish 1 shall be provided with seams and closures sealed by a suitable method. The film thickness shall be not less than 0.002 inches (2.0 mil plus or minus 20 percent).

5.1.2 Level B. The sodium sulfite or sodium bisulfite shall be packed as specified in 5.1.1, except that 100 pounds shall be packed in a close-fitting bag conforming to UU-B-48, grade 24 or 26 LW.

5.1.3 Level C. The sodium sulfite or sodium bisulfite shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The sulfites covered by this specification are both intended for the treatment of:

6.1.1 Boiler feedwater in steam-generating heating and power plants; and

6.1.2 Circulated high-temperature (300° - 500° F) water in pressurized heating systems.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type sulfite required (1.2)
(c) Unit quantities required (5.1).
(d) Levels of packing required (5.1).

6.3 Transportation description applicable to these items are: (a) Sodium sulfite anhydrous, and (b) Sodium bisulfite anhydrous. Carload minimum weight 40,000 pounds; truckload minimum weight, 36,000 pounds.

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